Direct Multi-Horizon GATv2 for Train Delay Prediction

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01 Introduction

- **Causes:** a **primary delay** spreads through shared infrastructure, dwell and headway constraints into **knock-on delays** [1].
- **The gap:** graph models forecast several steps by **autoregressive (AR) rollout** [2], feeding each prediction into the next step, so **errors accumulate** at later horizons.

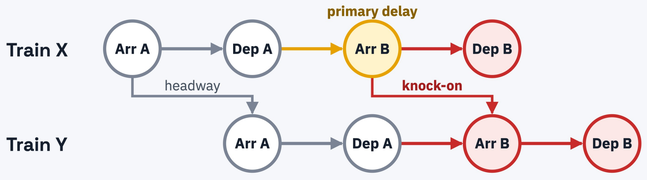


Figure 1. Delay propagation between two trains (after [2]).

My contribution

- **Direct Multi-Horizon (DMH) decoding:** from one GATv2 embedding of the current event, three heads predict the delay at **t+1, t+2 and t+3 at the same time**; no prediction is fed back.
- **Horizon-weighted aggregation:** every event is therefore predicted up to 3 times (from 1, 2 and 3 events earlier). These are merged into **one final delay**, weighting each horizon by 1 / its validation MAE (eq. 11–12).

Question: can DMH reduce error at later horizons?

02 Methodology

A · Data, event graph & problem

NYC subway observations (Subway Data NYC [4]) matched to the MTA static GTFS schedule. Each arrival and departure is a node.

- **Dwell** D = $t_{dep} - t_{arr}$ at the same stop
- **Run** R = $t_{arr} - t_{dep}$ of the previous stop
- **Headway** H = $t_i - t_{i-1}$, consecutive trains

Features: stop sequence, trip progress, stops left, time of day (sin/cos), rush hour.

Problem: forecast the next $K = 3$ events

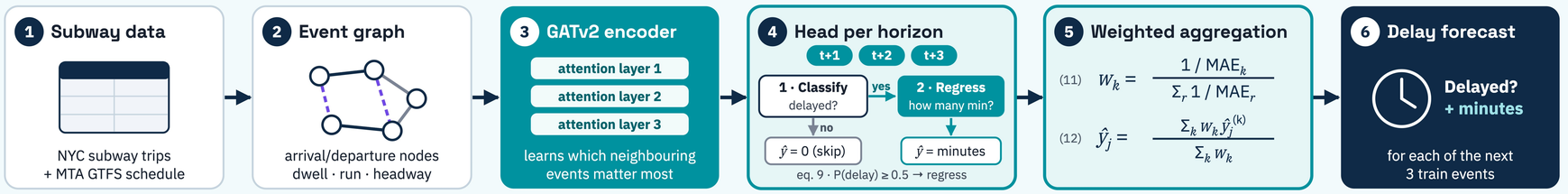
(3) $\mathbf{y}_i = [\, y_i^{(1)}, y_i^{(2)}, y_i^{(3)} \,]$ — target: observed delay at each of the next 3 events

(4) AR $\hat{y}_i^{(1)} \rightarrow \hat{y}_i^{(2)} \rightarrow \hat{y}_i^{(3)}$ — each step reuses the previous prediction, so **errors carry forward**

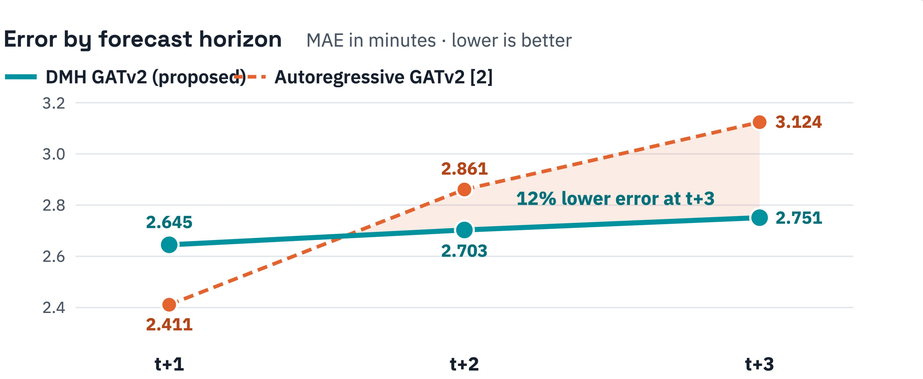
(5) DMH $[\, \hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \hat{y}_i^{(3)} \,]$ — all horizons at once from the graph embedding, **no feedback**

B My model: Direct Multi-Horizon GATv2 (DMH)

02 Methodology, continued



03 Results



+4%
error growth from t+1 to t+3 (autoregressive: +30%)

−12%
lower MAE at t+3 (DMH 2.75 vs AR 3.12 min)

94.9%
precision, best of all models

+3.9
recall points over autoregressive (93.8% vs 89.9%)

Table I · Model	MAE	RMSE	Accuracy	Precision	Recall	F1
Zero-Delay	6.7103	13.7107	0.5167	0.0000	0.0000	0.0000
Persistence	**0.7290**	**1.8926**	0.9290	0.9382	0.9132	0.9255
Autoregressive GATv2 [2]	3.1243	7.5966	0.9223	0.9380	0.8987	0.9179
DMH GATv2 + Aggregation (proposed)	2.5599	6.9135	**0.9458**	**0.9494**	**0.9378**	**0.9436**

Overall validation performance; bold = best. Persistence wins raw MAE because consecutive delays rarely jump.

04 Conclusion

- DMH curbs error accumulation: MAE grows only 4% from t+1 to t+3, versus 30% for autoregressive rollout.
- Best delay detection of all models (94.9% precision, 94.4% F1); persistence still wins raw MAE.
- **Next:** held-out test set, longer horizons, and networks with more sudden disruptions.

05 References

1. T. Spanninger, A. Trivella, B. Büchel, F. Corman. A review of train delay prediction approaches. J. Rail Transport Planning & Management 22, 2022.
2. V. D. A. Nguyen, Z. Li. Deep Learning to Identify the Spatio-Temporal Cascading Effects of Train Delays in a High-Density Network. GeoAI '25, 2025.
3. T. N. Kipf, M. Welling. Semi-Supervised Classification with Graph Convolutional Networks. ICLR 2017.
4. Subway Data NYC. subwaydata.nyc
5. S. Brody, U. Alon, E. Yahav. How Attentive Are Graph Attention Networks? ICLR 2022.